

SOP #9 – Troubleshooting Techniques for Emergency & Backup Generators

Applies to: Diesel, natural gas, or bi-fuel emergency generators in commercial/industrial settings

1. Common Failure Signs & Diagnoses

Symptom	Likely Cause	Recommended Action
Generator fails to start	Low battery voltage, starter failure	Test/replace battery, check starter relays
Engine cranks but no ignition	Fuel issue, glow plug/ignition fault	Check fuel filters, test ignition system
Unit trips during load test	Overload, faulty AVR	Reduce test load, calibrate voltage regulator
Excessive smoke (black/blue)	Dirty injectors, oil/fuel imbalance	Clean injectors, inspect PCV and breather
Oil or coolant leak	Gasket/seal degradation	Repair/reseal with OEM-approved components